

Original LCTRSs

LCTRS

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times int \Rightarrow State \\ \text{noncrit1} : Loc \\ \text{wait1} : Loc \\ \text{crit1} : Loc \\ \text{noncrit2} : Loc \\ \text{wait2} : Loc \\ \text{crit2} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{ll} 1 : & \text{state}(\text{noncrit1}, p2, y) \rightarrow \text{state}(\text{wait1}, p2, y) \quad [y > 0] \\ 2 : & \text{state}(\text{wait1}, p2, y) \rightarrow \text{state}(\text{crit1}, p2, y1) \quad [y1 = (y - 1)] \\ 3 : & \text{state}(\text{crit1}, p2, y) \rightarrow \text{state}(\text{noncrit1}, p2, y1) \quad [y1 = (y + 1)] \\ 4 : & \text{state}(p1, \text{noncrit2}, y) \rightarrow \text{state}(p1, \text{wait2}, y) \quad [y > 0] \\ 5 : & \text{state}(p1, \text{wait2}, y) \rightarrow \text{state}(p1, \text{crit2}, y1) \quad [y1 = (y - 1)] \\ 6 : & \text{state}(p1, \text{crit2}, y) \rightarrow \text{state}(p1, \text{noncrit2}, y1) \quad [y1 = (y + 1)] \\ 7 : & \text{state}(p1, p2, y) \rightarrow \text{success} \\ 8 : & \text{state}(\text{crit1}, \text{crit2}, y) \rightarrow \text{error} \end{array} \right\}$$

APR Problem

$$\{ \quad 9 : \quad \text{state}(\text{noncrit1}, \text{noncrit2}, 1) \Rightarrow \text{success} \quad \}$$

BV-LCTRS Version

LCTRS

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times bitvec_1 \Rightarrow State \\ \text{noncrit1} : Loc \\ \text{wait1} : Loc \\ \text{crit1} : Loc \\ \text{noncrit2} : Loc \\ \text{wait2} : Loc \\ \text{crit2} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{ll} 1 : & \text{state}(\text{noncrit1}, p2, y) \rightarrow \text{state}(\text{wait1}, p2, y) \quad [ybvugt\#b0] \\ 2 : & \text{state}(\text{wait1}, p2, y) \rightarrow \text{state}(\text{crit1}, p2, y1) \quad [y1 = (ybvs\#b1)] \\ 3 : & \text{state}(\text{crit1}, p2, y) \rightarrow \text{state}(\text{noncrit1}, p2, y1) \quad [y1 = (ybvadd\#b1)] \\ 4 : & \text{state}(p1, \text{noncrit2}, y) \rightarrow \text{state}(p1, \text{wait2}, y) \quad [ybvugt\#b0] \\ 5 : & \text{state}(p1, \text{wait2}, y) \rightarrow \text{state}(p1, \text{crit2}, y1) \quad [y1 = (ybvs\#b1)] \\ 6 : & \text{state}(p1, \text{crit2}, y) \rightarrow \text{state}(p1, \text{noncrit2}, y1) \quad [y1 = (ybvadd\#b1)] \\ 7 : & \text{state}(p1, p2, y) \rightarrow \text{success} \\ 8 : & \text{state}(\text{crit1}, \text{crit2}, y) \rightarrow \text{error} \end{array} \right\}$$

APR Problem

$$\{ \quad 9 : \quad \text{state}(\text{noncrit1}, \text{noncrit2}, \#b1) \Rightarrow \text{success} \quad \}$$